NUMBER RACK (REKENREK) ARITHMETIC STUDIO

Dual-Row 20-Bead 5-and-10 Grouping & Subitizing Frame

SERIES 1 • CARD 11 / 12 • DOMAIN: 5-AND-10 GROUPING & DUAL-ROW ARITHMETIC

CARD 11 / 12 • RACK-01

MATHEMATICAL GUIDE

1. CONCEPT & GEOMETRIC PROOF

5-AND-10 GROUPING & DUAL-ROW ARITHMETIC

Counting by ones is slow and error-prone. The Rekenrek organizes 20 beads into 5-anchors:
Row 1: 5 Red + 5 White • Row 2: 5 Red + 5 White
Sliding beads in groups turns math into subitizing!

CALIBRATED 102.0 MM SLIDER DATUM

- Beads: 10 Red + 10 White = 20 Beads (14.0 mm ⌀)
- Slider Rods: 2x 102.0 mm Octagonal Rods
- Upright Towers: 2x 49.0 mm Snap-Fit Towers (7U)

5-AND-10 REKENREK: 7 + 6 = 13



DOUBLES-PLUS-ONE

- Red: 5 + 5 = 10 (Instant T
- White: 2 + 1 = 3 Units

Sum = 10 + 3 = 13

Zero finger counting!

2. KIT PARTS INVENTORY (27 PIECES)

COMPONENT	QTY	SPEC	COLOR
Red Beads	10x	14.0 mm Bead	Crimson Red
White Beads	10x	14.0 mm Bead	Classic White
Slider Rods	2x	102.0 mm Rod	Matte Charcoal
Upright Towers	2x	49.0 mm	Matte Charcoal
Storage Tray	1x	Nesting Tray	Matte Charcoal
3x1 Storage Bin	1x	49.0 mm	Matte Charcoal
Stage Lid	1x	Slider Lid	Matte Charcoal

TOTAL: 20 Beads + 4 Frame Parts + Tray + Storage Bin + Lid = 27 Parts

DUAL-ROW 5-COLOR STRUCTURE

- 5 Red + 5 White per row: Instant 5-anchor visual subitizing.
- Both Rows Active: Models doubles, near-doubles, and bridging.
- Zero-Rattle Carry: Beads slide into nested tray slots.
- High Contrast: Crisp color separation across rows.

3. PACK-UP & SUPPORT

✓ 10-SECOND PACK-UP

Slide 10 red beads left and 10 white beads right on both rods. Place rods and towers into the nesting tray. Missing beads leave a visible gap. Snap lid shut.

✓ NO LOST PIECE GUARANTEE

Classrooms lose pieces. Every single math component is permanently stocked for single replacements.

stackcalc.io/parts
Order replacements anytime

✓ HANDS-ON TUTORIAL

Flip over for 5-step concrete lesson & live 3D Video QR!
Keep stored inside box lid!

FUNDAMENTAL THEOREM

Dual 5-and-10 grouping bridges perceptual subitizing directly into formal arithmetic operations.

STACKCALC ARCHIVAL INSTRUMENT LABS
Designed in Mountain View, California • Non-Toxic BPA-Free • Classroom Lab Stations: stackcalc.io/schools

OFFICIAL REFILL PORTAL
stackcalc.io/parts

LESSON GUIDE: DUAL-ROW BRIDGING & DOUBLES PLUS ONE

Step-by-Step Hands-On Tutorial: 7 + 6 = 13 via 5-and-5 Doubles

SERIES 1 • CARD 11 OF 12 • DOMAIN: 5-AND-10 GROUPING & DUAL-ROW ARITHMETIC

CARD 11 / 12 • RACK-01

LESSON TUTORIAL

SCAN FOR 3D VIDEO DEMO

OPEN CAMERA & SCAN

stackcalc.io/products/number-rack/

Zero app install • iPad & Chromebook

CURRICULAR ALIGNMENT

- CCSS 1.OA.C.6 (Addition Fluency)
- CCSS 2.OA.B.2 (Mental Math)
- TEKS 111.3.b.3.C (Compose/Deco...)

Grades K–3 Subitizing & Addition

GUIDING INQUIRY

When you slide 7 beads on top and 6 on bottom, how many red beads are on the left side altogether?

1 STEP 1: SLIDE 7 BEADS ON TOP ROW

Push all beads to the right side of the rack (resting zero).
Slide 7 beads to the left on the top rod in one push:
All 5 red beads + 2 white beads = 7 beads!
Notice: 5-anchor lets you see 7 instantly

2 STEP 2: SLIDE 6 BEADS ON BOTTOM ROW

Slide 6 beads to the left on the bottom rod in one push:
All 5 red beads + 1 white bead = 6 beads!
We want to find 7 + 6 without counting one-by-one.

3–4 STEPS 3 & 4: GROUP DOUBLES & REGROUP

3. GROUP RED BEADS: Look at the red beads on the left: 5 + 5 = 10 (Instant Ten)!
4. ADD WHITE BEADS: Look at the white beads: 2 + 1 = 3 units.
Total: 10 + 3 = 13! Therefore: 7 + 6 = 13!
Calculator Verification: Press Number Rack Subitizing Icon ([▦]) and run [7] [ENTER] [6] [+] → 13.0000.

5 CORE PRINCIPLE: DOUBLES MAKE MENTAL MATH EFFORTLESS

Instead of counting 6 fingers past 7, the Rekenrek decomposes 7 + 6 into (5 + 2) + (5 + 1). The two groups of 5 red beads instantly make 10, leaving 3 white beads. Addition becomes instantaneous visual pattern recognition rather than tedious memorization!

★ EXTENSION INVESTIGATION: SOLVE: 8 + 7 USING 5-AND-5

Slide 8 beads on top (5 red + 3 white) and 7 on bottom (5 red + 2 white). Group the 10 red beads first! What is the sum? (Scan QR for 3D animated solution!)

→ Assemble with physical blocks, then scan QR code for 3D animated solution.